

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458220.6518 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.183 +/- 0.014
Transit depth (Rp/Rs)^2	3.36 +/- 0.52 [%]
Orbital Inclination (inc)	82.75 +/- 1.74
Airmass coefficient 1 (a1)	0.7079 +/- 0.0076
Airmass coefficient 2 (a2)	0.26 +/- 0.1
Scatter in the residuals of the lightcurve fit is	1.09 %
Best Comparison Star	#3 - [364, 224]
Optimal Aperture	3.32
Optimal Annulus	7.35
Transit Duration (day)	0.0463 +/- 0.0028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.183 ± 0.014 vs 0.1594 (deviation 1.69σ)
Duration fit vs published	0.0463 d vs 0.0512 d (deviation 1.77σ)
Mid-time O–C	-3.7 min
Depth SNR	13.1
Residual scatter	1.09 %

Light curve

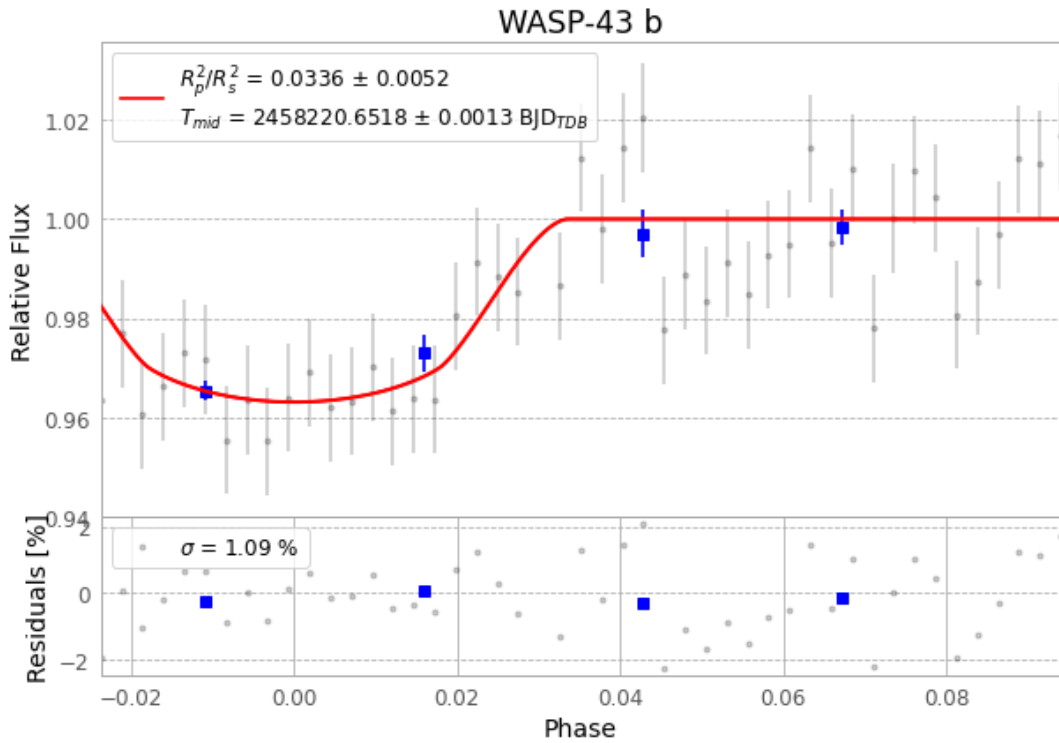


Figure 1 — FinalLightCurve_WASP-43 b_2018-04-11.png (SHA-256 37afb665fc2407dd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [431, 226], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6e7e181aba9c3205...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

47 FITS frames (31 MB), WASP-43180412030312.FITS → WASP-43180412052112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-180412.log (805ba695e343...), FinalLightCurve_WASP-43 b_2018-04-11.png (37afb665fc24...), FinalLightCurve_WASP-43 b_2018-04-11.csv (d39e4cf4b915...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-24 22:12Z.